



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2023 - 2024)

JEE(Advanced)
PRE-SCORE
05-11-2023

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : TOAS, TNAS, TAAS & TTAS)

Time : 3 Hours

PAPER-1 (OPTIONAL)

Maximum Marks : 180

IMPORTANT NOTE : Students having 8 digits **Form No.** must fill two zero before their Form No. in OMR. For example, if your **Form No.** is 12345678, then you have to fill **0012345678**.

READ THE INSTRUCTIONS CAREFULLY

GENERAL :

1. This sealed booklet is your Question Paper. Do not break the seal till you are told to do so.
2. Use the Optical Response sheet (ORS) provided separately for answering the questions.
3. Blank spaces are provided within this booklet for rough work.
4. Write your name, form number and sign in the space provided on the back cover of this booklet.
5. After breaking the seal of the booklet, verify that the booklet contains **24** pages and that all the **19** questions in each subject and along with the options are legible. If not, contact the invigilator for replacement of the booklet.
6. You are allowed to take away the Question Paper at the end of the examination.

OPTICAL RESPONSE SHEET :

7. The ORS will be collected by the invigilator at the end of the examination.
8. Do not tamper with or mutilate the ORS. **Do not use the ORS for rough work.**
9. Write your name, form number and sign with pen in the space provided for this purpose on the ORS. **Do not write any of these details anywhere else on the ORS.** Darken the appropriate bubble under each digit of your form number.

DARKENING THE BUBBLES ON THE ORS :

10. Use a **BLACK BALL POINT PEN** to darken the bubbles on the ORS.
11. Darken the bubble  **COMPLETELY.**
12. The correct way of darkening a bubble is as : 
13. The ORS is machine-gradable. Ensure that the bubbles are darkened in the correct way.
14. Darken the bubbles **ONLY IF** you are sure of the answer. There is **NO WAY** to erase or "un-darken" a darkened bubble.
15. Take **$g = 10 \text{ m/s}^2$** unless otherwise stated.

QUESTION PAPER FORMAT :

16. The question paper has three parts : Physics, Chemistry and Mathematics.

Please see the last page of this booklet for rest of the instructions

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

SOME USEFUL CONSTANTS

Atomic No. : H = 1, B = 5, C = 6, N = 7, O = 8, F = 9, Al = 13, P = 15, S = 16, Cl = 17, Br = 35, Xe = 54, Ce = 58

Atomic masses : H = 1, Li = 7, B = 11, C = 12, N = 14, O = 16, F = 19, Na = 23, Mg = 24, Al = 27, P = 31, S = 32, Cl = 35.5, Ca = 40, Fe = 56, Br = 80, I = 127, Xe = 131, Ba = 137, Ce = 140

- **Boltzmann constant** $k = 1.38 \times 10^{-23} \text{ JK}^{-1}$
- **Coulomb's law constant** $\frac{1}{4\pi\epsilon_0} = 9 \times 10^9$
- **Universal gravitational constant** $G = 6.67259 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$
- **Speed of light in vacuum** $c = 3 \times 10^8 \text{ ms}^{-1}$
- **Stefan-Boltzmann constant** $\sigma = 5.67 \times 10^{-8} \text{ Wm}^{-2}\text{-K}^{-4}$
- **Wien's displacement law constant** $b = 2.89 \times 10^{-3} \text{ m-K}$
- **Permeability of vacuum** $\mu_0 = 4\pi \times 10^{-7} \text{ NA}^{-2}$
- **Permittivity of vacuum** $\epsilon_0 = \frac{1}{\mu_0 c^2}$
- **Planck constant** $h = 6.63 \times 10^{-34} \text{ J-s}$

Space for Rough Work

BEWARE OF NEGATIVE MARKING

PART-1 : PHYSICS

SECTION-I (i) : (Maximum Marks: 12)

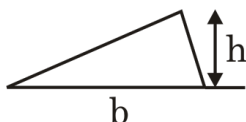
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If ONLY the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. If we find the moment of inertia of a scalene triangular plate ABC about its base as shown. The mass of the plate is M. The base length is b and altitude is h. Moment of inertia is :-



$$(A) \frac{Mh^2}{12}$$

$$(B) \quad \frac{Mh^2}{3}$$

$$(C) \quad M \frac{h^2 + b^2}{12}$$

$$(D) \frac{Mh^2}{6}$$

2. Assume that earth has a mass of 6×10^{24} kg and a radius of 6400 km. Earth has a population of 8 billion people each of mass 50 kg. On the call of our prime minister, the entire population of earth lines up on the equator and move with a velocity of 1 m/s relative to the earth from west to east. What is the change in the time period of revolution of earth?

(A) Approx. 10^{-15} sec

(B) Approx. 10^{-10} sec

(C) Approx. 10^{-12} sec

(D) Approx. 10^{-17} sec

3. In figure 1, we see a ball C colliding against a ball B which is connected to a ball A by a loose inextensible string. There is no friction anywhere. In figure 2, the ball C collides against ball B which is connected to ball A by a taut inextensible string. If the collision is elastic in each of the cases,

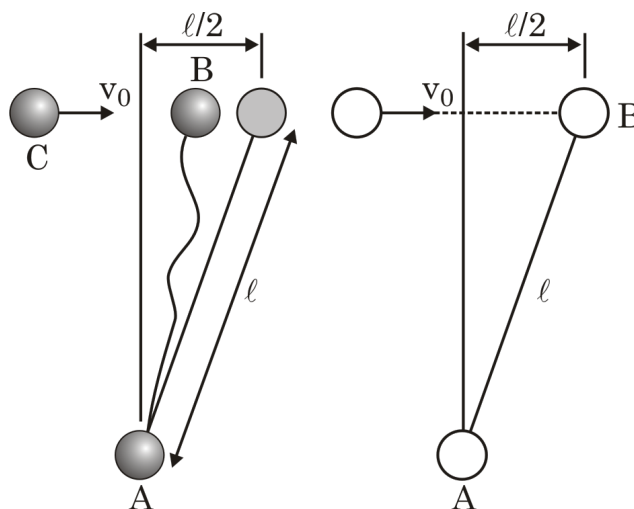
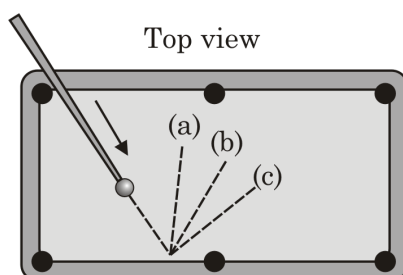


Fig.(1)

Fig.(2)

- (A) The mechanical energy is conserved during and after the collision in both the cases.
- (B) The mechanical energy is conserved during and after the collision in case 1 but not in 2.
- (C) The mechanical energy is conserved during and after the collision in case 2 but not in 1.
- (D) The mechanical energy is not conserved during and after the collision in both the cases.
4. An experienced pool player can control the placement of a ball after it hits an edge bumper by putting spin, or “English,” on the ball. This is accomplished by aiming the cue stick so that it hits the ball either left or right of the ball’s centre. Suppose a player gives a ball English by hitting it to the left of centre (left from his perspective). Which path in Figure does the ball take after it collides with the bumper? How is it rotating before collision with the bumper as seen from the top?



- (A) Path a with clockwise spin.
- (B) Path b with clockwise spin.
- (C) Path c with anticlockwise spin.
- (D) Path a with anticlockwise spin.

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

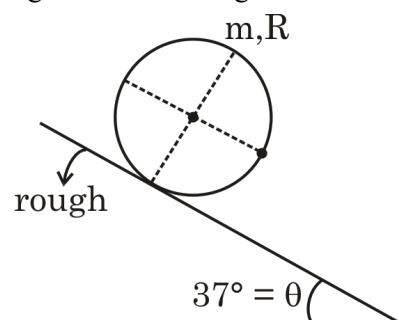
Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

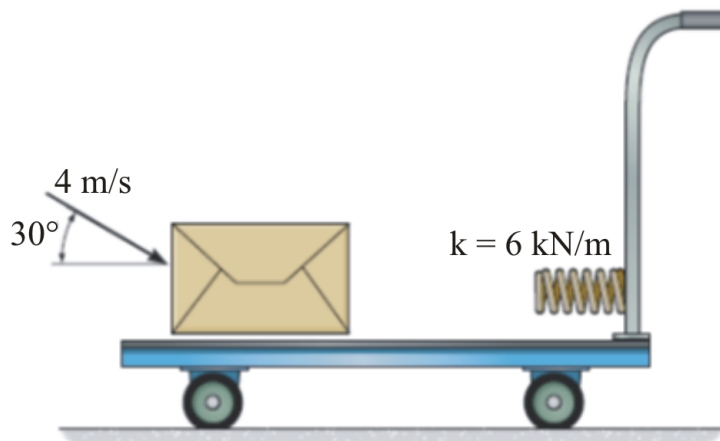
- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. A uniform circular ring of mass $m = 1\text{ kg}$ and radius $R = 20\text{ cm}$ is placed on a rough inclined plane of inclination $\theta = 37^\circ$ as shown in the figure. Then choose the correct option(s) just after the ring is released from rest. (there is no slipping between the ring and inclined plane and $g = 10\text{ m/s}^2$)

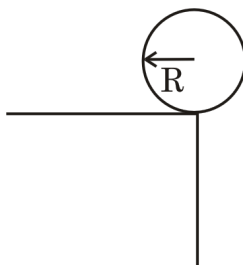


- (A) The frictional force acting on the ring due to inclined plane is 4N.
- (B) The frictional force acting on the ring due to inclined plane is 3N.
- (C) The normal force acting on the ring due to inclined plane is 8 N.
- (D) The normal force acting on the ring due to inclined plane is 11N.

6. The 40-kg package is thrown with a speed of 4 m/s onto the cart having a mass of 20 kg. If it slides on the smooth surface and strikes the spring, determine the velocity u_c of the cart at the instant the package fully compresses the spring. What is the maximum compression x of the spring? Neglect rolling resistance of the cart. What is the velocity v_p of the package after the spring comes back to the natural length? What is the velocity v_c of the cart at that instant?

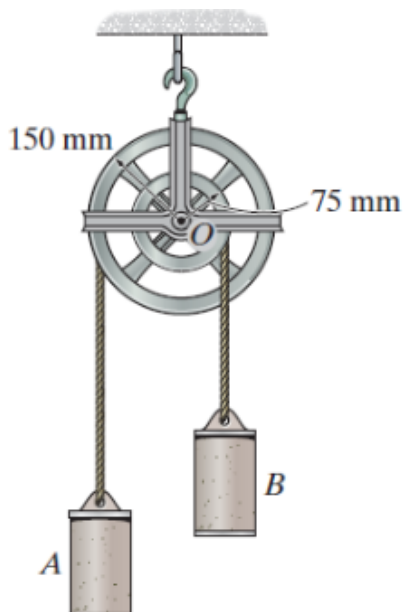


- (A) $u_c = \frac{4}{\sqrt{3}} \text{ m/s}$ (B) $v_c = \frac{8}{\sqrt{3}} \text{ m/s}$
 (C) $v_p = \frac{2}{\sqrt{3}} \text{ m/s}$ (D) $x = \sqrt{\frac{2}{75}} \text{ m}$
7. A small uniform ring of mass m and radius R is held at the corner of a table and released. It was in unstable equilibrium and so it falls off to the ground below. We have two options for the table, both having the same height. The first table has a rough edge so that the ring does not slip when it is in contact with the edge, and the second table has a smooth edge.

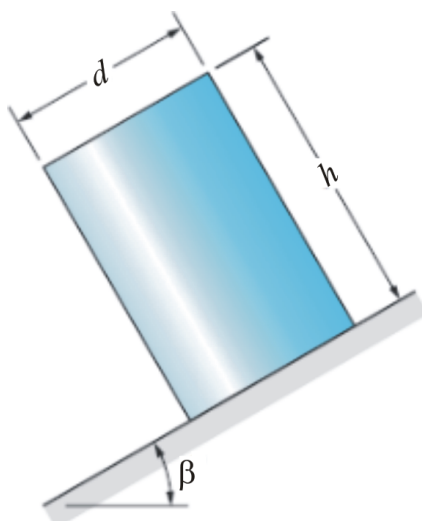


- (A) In the case of the first table, the ring rotates by a larger angle before leaving contact with the edge.
 (B) In the case of first table, the angular velocity of the ring remains constant during its motion in the air.
 (C) In the case of second table, the ring travels with lesser horizontal velocity during its motion in the air.
 (D) In the case of second table, angular velocity of ring in air is more than ring first table.

8. If the system is released from rest, determine the speed of the 20-kg cylinders A and B after A has moved by a distance of 2 m. The differential pulley has a mass of 15 kg with a radius of gyration about its centre of mass $k = 100$ mm.

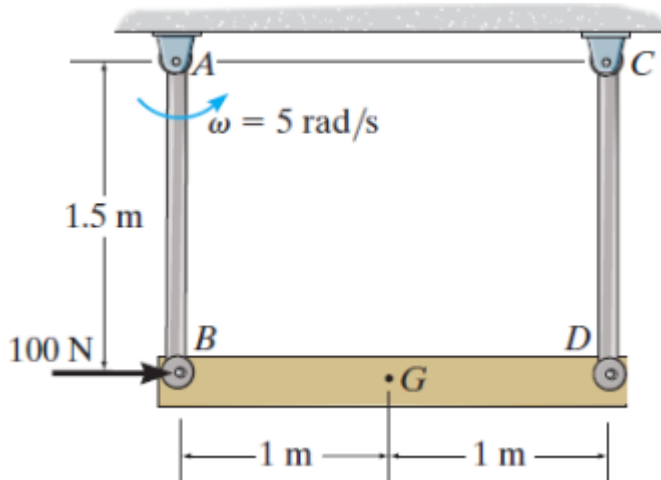


- (A) A will move up. (B) B will move up.
- (C) A will have a speed of $\frac{4\sqrt{15}}{5}$ m/s. (D) B will have a speed of $\frac{30}{\sqrt{2445}}$ m/s.
9. A uniform cylinder of mass M , height h and diameter d is kept on a rough inclined plane. The coefficient of static friction and kinetic friction are both μ . Select the correct statement(s).



- (A) If $\mu > \tan \beta$ and $\tan \beta > d/h$ Cylinder will neither topple nor slide.
- (B) If $\mu > \tan \beta$ and $\tan \beta > d/h$ Cylinder will topple but will not slide.
- (C) If $\mu < \tan \beta$ and $\tan \beta > d/h$ Cylinder will slide but not topple.
- (D) If $\mu < \tan \beta$ and $\tan \beta < d/h$ Cylinder will slide but not topple.

10. At the instant shown both rods of negligible mass swing with a counterclockwise angular velocity of 5 rad/s while the 50 kg bar is subjected to the 100 N horizontal force. Determine the tension developed in the rods and the angular acceleration of the rods at this instant.



- (A) The angular acceleration is anti-clockwise.
 (B) The angular acceleration is $4/3 \text{ rad/s}^2$.
 (C) Tension is approx. 1.18 kN .
 (D) Tension is more in rod AB than in CD.

SECTION-II (i) : (Maximum Marks: 12)

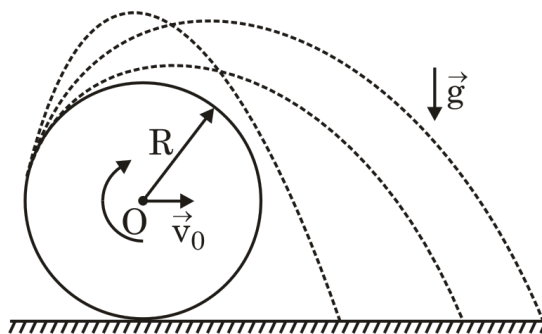
- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

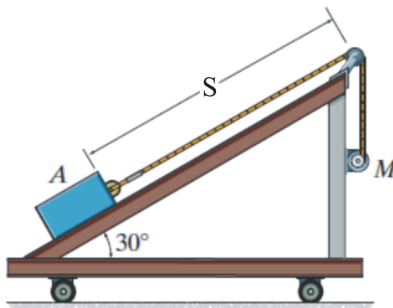
A wheel of radius $R = 10.0$ cm moves without slippage along horizontal surface with a constant velocity $v_0 = 2.0$ m/s.



1. Determine maximum height H (in cm) that can be reached by a splash of mud, which is separated from that moving wheel. Assume that friction with air can be neglected, while height H is measured from the horizontal surface. For calculations use acceleration due to gravity as 10 m/s^2 .
2. Find the maximum height (in cm) from the ground reached by the splash of mud that just grazes the moving wheel again when it is travelling downwards.

Paragraph for Questions 3 and 4

The 20-kg block A is towed up the ramp of the 40-kg cart using the motor M mounted on the side of the cart. The motor begins winding in the cable with a constant velocity of 5m/s, measured relative to the cart. Both the block and cart are at rest initially. The kinetic friction between the block and the ramp is absent. Neglect rolling resistance.



3. Determine how far (in m) the cart will move when the block has travelled a distance 2 m up the ramp.
4. Determine the work done by the motor (in J) till then.

Paragraph for Questions 5 and 6

A uniform rod of mass 100 g and length 50.0 cm rotates in a horizontal plane about a fixed, vertical, frictionless pin through its centre. Two small beads, each of mass 30.0 g, are mounted on the rod so that they are able to slide without friction along its length. Initially the beads are held by catches at positions 10.0 cm on each side of centre, at which time the system rotates at an angular speed of 20.0 rad/s. Suddenly, the catches are released and the small beads slide outward along the rod.

5. Find the angular speed of the system (in rad/s) at the instant the beads reach the ends of the rod.
6. If the beads fly off the ends, What is the speed (in m/s) of each of the beads just after that? Fill v^2 .

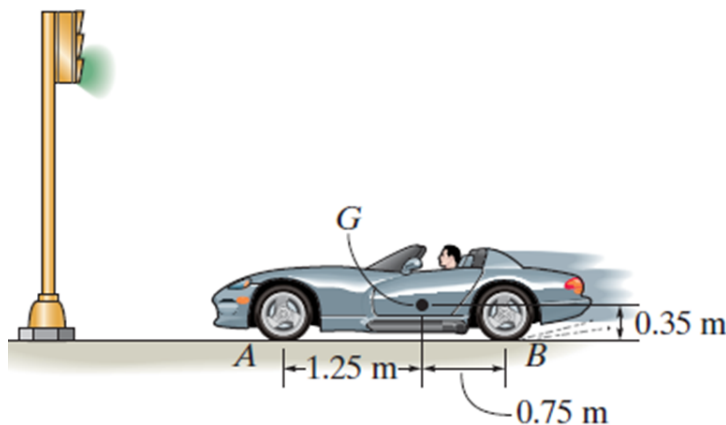
SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

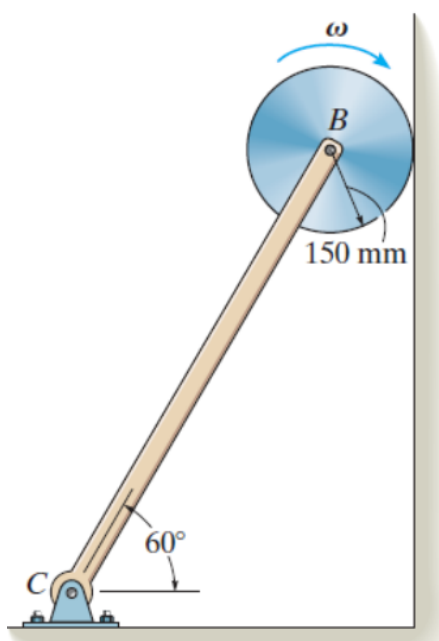
Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

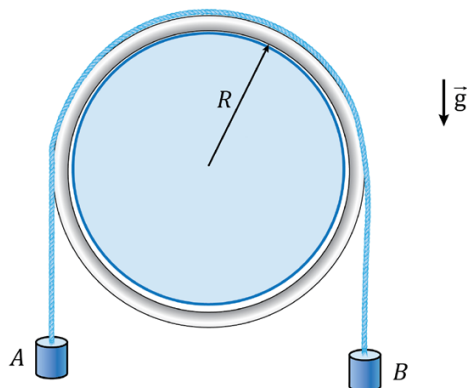
7. The sports car has a mass of 1500 kg and a centre of mass at G. Determine the shortest time (in sec) it takes for it to reach a speed of 108 km/hr, starting from rest, if the engine only drives the rear wheels, whereas the front wheels are free rolling. The coefficient of static friction between the wheels and the road is 0.2. Neglect the mass of the wheels for the calculation. Round off to nearest integer.



8. The disk has a mass of 20 kg and is originally spinning at the end of the massless rod with an angular velocity of $1000(2 - \sqrt{3})$ rad/s. If it is then placed against the wall, where the coefficient of kinetic friction is $2 - \sqrt{3}$, determine the time (in sec) required for the motion to stop.



9. A ring with uniformly distributed mass m hangs at the fixed horizontal cylinder, which has radius slightly less than inner radius of the ring R . Coefficient of friction between the ring and fixed cylinder is $\mu = 0.1$. On top of the ring is thrown a non-stretchable rope with negligible mass, where weights with masses $m_A = 2m$ and $m_B = 5m$ are attached to the ends of the rope.



Determine acceleration (in m/s^2) a of the weight B after release of the system in the gravity of the Earth. For calculations use acceleration due to gravity as $g = 10 \text{ m/s}^2$. Assume that rope does not slip on the ring. Round off to nearest integer.

PART-2 : CHEMISTRY
SECTION-I (i) : (Maximum Marks: 12)

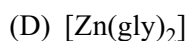
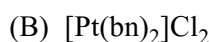
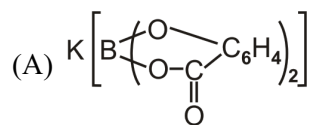
- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

1. Which of the following species cannot be resolved into 'd' and 'ℓ' form.



2. Sodium lauryl sulphate is widely used as a collector in froth flotation process. It can be written as $\text{C}_{12}\text{H}_{25}\text{OSO}_3\text{Na}$. Which is correct regarding its function as a collector particle during froth floatation.

(A) Its hydrophilic end gets attached to sulphide ore particle which then gets attached to air bubble

(B) Its hydrophobic end tends to stay in water thereby facilitating ore particle movement towards surface

(C) It stays at air-water interface and the ore settles at the bottom

(D) It's hydrophobic end gets attached to the ore particle and thus enhances the floating characteristics

3. A colourless odourless gas "X" can be prepared by action of steam over hot coke on a surface of catalyst at high temperature. The same gas "X" reduces 2-ethylantraquinone in the presence of Ni/Pd catalyst. Which of the following is **CORRECT** for gas "X".

(A) Bond strength of $\text{F}_2 >$ Bond strength of X

(B) "X" is obtained at cathode in DOWN's cell

(C) "X" boils at lower temperature as compared to lightest noble gas

(D) "X" react with iodine only in the presence of catalyst.

4. Which of the following metals involve oxidation of metal followed by reduction during their extraction.

(A) Ag from Ag_2S

(B) Fe from Fe_2O_3

(C) Au from Auriferous rocks

(D) Sn from SnO_2

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

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- **For Example :** If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

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5. Metal "M" can be reduced from its oxide via carbon reduction. It produces H_2 on dissolving metal in aqueous alkali. The complex $MCl_2 \cdot 4NH_3$ produces white precipitate with $AgNO_3$. If M and M^{2+} are diamagnetic in gaseous state, then
- (A) "M" is refined by distillation
(B) "M" is refined by liquation
(C) Electrowinning of "M" is done using aqueous electrolyte
(D) One of the ore of M can be concentrated by froth floatation
6. Which of the following complex specie(s) are paramagnetic & low spin.
- (A) $[NiF_6]^{2-}$ (B) $[Fe(CN)_6]^{3-}$
(C) $[Co(CN)_6]^{4-}$ (D) $[Cr(CN)_6]^{3-}$
7. Increase in concentration of which specie(s) will cause breakdown of ozone in atmosphere.
- (A) $\dot{C}l$ (B) $\dot{C}lO$ (C) O_2 (D) NO_2ClO

8. "X" is used as solvent for dry cleaning and is carcinogenic in nature. "Y" is used as a bleaching agent in process of laundry. "X" is planar "Y" is non-planar.
Which is/are correct for "X" & "Y".
- (A) Both "X" & "Y" are non-polar
(B) "X" is non polar & "Y" is polar
(C) Urea acts as a stabilizer to store "Y"
(D) Liquified CO_2 can be used for dry-cleaning in place of "X"
9. Which is/are **CORRECT**.
- (A) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ contains hydrogen bonded H_2O .
(B) $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$ is interstitial water
(C) $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ has six H_2O are coordinated with metal
(D) Fluorine reacts with NaOH(aq.) to produce OF_2
10. In given four options, each option contains a complex salt which have cationic as well as anionic part is complex sphere. Which of the following coordination compound show co-ordination isomerism as well as stereo isomerism.
- (A) $[\text{Pt(en)}_2] [\text{PtCl}_4]$
(B) $[\text{Co(NH}_3)_6] [\text{Co(CN)}_4(\text{NO}_2)_2]$
(C) $[\text{Pt(en)}_2\text{Br}_2] [\text{PtCl}_4]$
(D) $[\text{Ag(NH}_3)_2] [\text{Ag(CN)}_2]$

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
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- Answer to each question will be evaluated according to the following marking scheme:

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Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Effective atomic number provides a convenient way to compare stability of carbonyl complexes. It is related to Sidgwick's EAN rule which compared complex compounds using analogy of inert gases.

1. The sum of magnetic moments of the complexes that do not follow Sidgwick's EAN rule among $[\text{Fe}(\text{CO})_5]$, $\text{K}_2[\text{PtCl}_6]$, $[\text{Fe}(\text{NH}_3)_6]\text{Cl}_2$, $[\text{Ni}(\text{NH}_3)_6]\text{SO}_4$ is _____.
2. Number of complex compounds from given below which have more number of valence electrons in the orbital of each central metal as compared to the number of valence electrons on central metal in $[\text{Mn}(\text{CO})_5]$ complex.
 $\text{Mn}_2(\text{CO})_{10}$, $[\text{Fe}(\text{CO})_5]$, $[\text{Ni}(\text{CO})_4]$, $\text{Co}_2(\text{CO})_8$, $\text{Fe}_2(\text{CO})_9$

Paragraph for Questions 3 and 4

Leaching is a chemical method for concentration of ore involving a leaching agent and other reagent. Usually ores of Au, Al, Ag, Cu are concentrated by this method.

3. During leaching of Bauxite by Serpeck's method, two gaseous compounds are formed. The sum of oxidation states of all the least electronegative atom(s) in both compounds is _____.
4. Leaching of Ag in dilute KCN in presence of O_2 produces a compound X. Sum of coordination number of X and EAN of X is _____.

Paragraph for Questions 5 and 6

Tropospheric pollution occurs due to presence of undesirable solids or gaseous particles in air. These pollutants are divided into two categories viz. gaseous pollutants and particulate pollutants.

5. Number of reaction(s) that yield a gas which can act as gaseous air pollutant.
 - (i) $\text{NH}_3 + \text{O}_2 \xrightarrow[500\text{K, 9 bar}]{\text{Pt-Rh catalyst}}$
 - (ii) $\text{NaSO}_3 \xrightarrow{\text{Zn} + \text{dil. H}_2\text{SO}_4}$
 - (iii) $\text{NaNO}_2 \xrightarrow{\text{dil. HCl}}$
 - (iv) $\text{HCOOH} \xrightarrow{\text{conc. H}_2\text{SO}_4}$
6. To create a brown haze in an artificial atmosphere, 5 mole NO_2 is radiated with energy equivalent to that of sunlight to break NO_2 molecule into NO & $[\text{O}]$ atom.
 Sufficient O_2 is supplied for formation of O_3 by reaction of O_2 and $[\text{O}]$. If percentage yield of dissociation of NO_2 is 80%, then calculate amount of NO_2 finally present in artificial atmosphere (in moles)

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

-
7. A sample of hard water which contain equimolar mixture of $\text{Ca}(\text{HCO}_3)_2$ and $\text{Mg}(\text{HCO}_3)_2$, is treated with 3000 mL, 0.5 M $\text{Ca}(\text{OH})_2$ solution such that none of the reactants are left. Amount of precipitate formed (in grams) will be _____.
(Consider all Mg^{2+} to be precipitated as $\text{Mg}(\text{OH})_2$ and all Ca^{2+} as CaCO_3)
(Ca = 40, C = 12, Mg = 24, O = 16, H = 1)
8. Number of linkage isomers of $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$ will be :-
9. Number of process(es) that involve no chemical change
- (i) Distillation of Zn
 - (ii) Mond's process
 - (iii) Liquation
 - (iv) Poling
 - (v) Zone refining
 - (vi) Thermal decomposition
 - (vii) Mac-Arthur forest process
 - (viii) Cupellation
 - (ix) Amalgamation
 - (x) Parke's method

PART-3 : MATHEMATICS
SECTION-I (i) : (Maximum Marks: 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks : +3 If **ONLY** the correct option is chosen.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered)

Negative Marks : -1 In all other cases

- Let M_n be the $n \times n$ matrix with entries as follow for $1 \leq i \leq n$, $m_{i,i} = 10$; for $1 \leq i \leq n-1$, $m_{i+1,i} = m_{i,i+1} = 3$ all other entries in M_n are zero. Let D_n be the determinant of matrix M_n , then $\sum_{n=1}^{\infty} \frac{1}{8D_n + 1}$
 (A) $\frac{1}{4}$ (B) $\frac{1}{8}$ (C) $\frac{1}{36}$ (D) $\frac{1}{72}$
- Let ω be a complex cube root of unity with $\omega \neq 1$ and $P = [p_{ij}]$ be a $n \times n$ matrix with $p_{ij} = \omega^{i+j}$. Then $P^2 = 0$, when $n =$
 (A) 57 (B) 55 (C) 58 (D) 56
- Evaluate $\sum_{k=0}^{n+1} \sum_{j=1}^{k+1} \sum_{i=1}^k {}^iC_{j-1}$ where $n > 2$ and ${}^nC_r = 0 \forall r > n$
 (A) $2^{n+3} - 2n - 6$
 (B) $2^{n+3} - 2n - 4$
 (C) $2^{n+2} - 4n - 2$
 (D) None of these
- Let x be a fixed real number which is not an integer multiple of π . For each natural number n let $A_n = [a_{jk}]$ be the $n \times n$ -matrix defined by

$$a_{jk} = \begin{cases} 0 & ; \text{ for } |j-k| > 1 \\ 1 & ; \text{ for } |j-k| = 1 \\ 2 \cos x & ; \text{ for } j = k \end{cases}$$
 If $D_n = \det A_n$ then the solution of $D_3 - 2D_2 \cos x + D_1 = 0$ is $[0, 2\pi]$ is
 (A) infinitely many (B) 0 (C) 2 (D) 3

SECTION-I (ii) : (Maximum Marks: 24)

- This section contains **SIX (06)** questions.
- Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen.

Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen and both of which are correct.

Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : -2 In all other cases.

- **For Example** : If first, third and fourth are the **ONLY** three correct options for a question with second option being an incorrect option; selecting only all the three correct options will result in +4 marks. Selecting only two of the three correct options (e.g. the first and fourth options), without selecting any incorrect option (second option in this case), will result in +2 marks. Selecting only one of the three correct options (either first or third or fourth option), without selecting any incorrect option (second option in this case), will result in +1 marks. Selecting any incorrect option(s) (second option in this case), with or without selection of any correct option(s) will result in -2 marks.

5. The number of ways to paint a regular octahedron (it has eight equilateral triangular faces, 6 vertices, 12 sides) with 8 different colors is (All colors to be used)
 - (A) Same as the indistinguishable number of ways to name the vertices of a cube with letters A, B, C, D, E, F, G, H
 - (B) having sum of digits 15
 - (C) having 32 positive divisors
 - (D) having four prime factors
6. If the elements of a 2×2 matrix A are real positive and distinct such that $\det(A + A^T) = 0$ where $\det(A)$ represent determinant value of matrix A, then :
 - (A) $\det A > 0$
 - (B) $\det A \geq 0$
 - (C) $\det(A - A^T) > 0$
 - (D) $\det(AA^T) > 0$
7. If A is non-singular matrix of order $n \times n$ such that $3ABA^{-1} + A = 2A^{-1}BA$, then :
 - (A) A and B both are identity matrices
 - (B) $|A + B| = 0$
 - (C) $|ABA^{-1} - A^{-1}BA| = 0$
 - (D) A + B is not a singular matrix

8. Let 'n' be an integer with $n \geq 3$. Let $P_1P_2P_3 \dots P_n$ be a regular n – sided polygon inscribed in a circle. Three points P_i , P_j and P_k are randomly chosen, where i, j and k are distinct integers from 1 to n. The number of triangles such that the triangle $P_iP_jP_k$ is obtuse is

(A) $\frac{n(n-2)(n-4)}{8}$, if n is even

(B) $\frac{n(n-1)(n-3)}{8}$, if n is odd

(C) $\frac{n(n-1)(n-2)}{6}$, if n is even

(D) $\frac{n(n+1)(n+2)}{6}$, if n is odd

9. Let N be the number of ways to place the integers 1 through 12 in the 12 cells of a 2×6 grid so that for any two cells sharing a side, the difference between the numbers in those cells is not divisible by 3. One way to do this is shown below.

1	3	5	7	9	11
2	4	6	8	10	12

(A) $N = 2^{11} \cdot 3^5 \cdot 5$

(B) $N = 2^{11} \cdot 3^5 \cdot 5^2$

(C) Total divisor of $N = 144$

(D) Total divisor of $N = 216$

10. If value of $3P = {}^{30}C_5 - {}^{30}C_6 + {}^{30}C_7 - \dots - {}^{30}C_{28} + 29$, and value of ${}^QC_R = {}^{100}C_6 + 4 {}^{100}C_7 + 6 {}^{100}C_8 + 4 {}^{100}C_9 + {}^{100}C_{10}$, then

(A) number of positive divisors of P are 16

(B) $Q + R$ can be divisible either by 9 or 19

(C) if $abcd = P$, $a, b, c, d \in I^+$, then possible number of order pairs of (a, b, c, d) are 4^4

(D) P is an even integer

SECTION-II (i) : (Maximum Marks: 12)

- This section contains **THREE (03)** question stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value corresponding to the answer in the designated place.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +2 If ONLY the correct numerical value is entered at the designated place;

Zero Marks : 0 In all other cases.

Paragraph for Questions 1 and 2

Consider two 3×3 matrices A and B satisfying $A = \text{adj}(B) - B^T$ and $B = \text{adj}(A) - A^T$. Also given that A is a non singular matrix. (Where P^T denotes transpose of matrix P).

1. $|A| + |B|$ is equal to :
2. $|AB + BA|$ is equal to :

Paragraph for Questions 3 and 4

If $(1 + px + x^2)^n = 1 + a_1x + a_2x^2 + \dots + a_{2n}x^{2n}$, ($n \in \mathbb{N}$) then

3. The remainder obtained, when $a_1 + 5a_2 + 9a_3 + 7a_4 + \dots + (8n - 3)a_{2n}$, is divided by $(p + 2)$ is :
4. The value of $a_1 + 3a_2 + 5a_3 + 7a_4 + \dots + (4n - 1)a_{2n}$, when $p = -3$ and $n = 4$ is :

Paragraph for Questions 5 and 6

A triangle is called an integral triangle if all its sides are integers. If a, b, and c are sides of an integral triangle, then we can assume that $a \leq b \leq c$, because any other permutation of a, b, and c will yield the same result. Now, answer the following question.

5. The number of integral isosceles or equilateral triangle's, none of whose side exceeds 4, is :
6. The number of integral isosceles or equilateral triangles, none of whose side exceeds 6 is :

SECTION-II (ii) : (Maximum Marks: 12)

- This section contains **THREE (03)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**
- For each question, enter the correct integer value of the answer in the place designated to enter the answer.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +4 If only the correct answer is given.

Zero Marks : 0 In all other cases

7. Ram chooses five subsets S_1, S_2, S_3, S_4, S_5 of $\{1, 2, 3, 4, 5, 6, 7\}$ such that $|S_1| = 1, |S_2| = 2, |S_3| = 3, |S_4| = 4$, and $|S_5| = 5$. Moreover, for all $1 \leq i < j \leq 5$, either $S_i \cap S_j = S_i$ or $S_i \cap S_j = \phi$ (in other words, the intersection of S_i and S_j is either S_i or the empty set). If Ram can achieve this in λ ways then the 3rd digit of λ from beginning is _____
8. If $p = \sum_{r=1}^{50} \left\{ \sum_{k=1}^r \left(\frac{(-1)^{r-1}}{k} \cdot {}^{50}C_r \right) \right\}$ then the value of $200(p)$ equals
9. Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where a, b, c, d are positive integers arranged in ascending order and exactly two of a, b, c, d are prime numbers and also pair wise coprime and satisfies $A^2 - 16A - 17I = 0$ (where I is an identity matrix of order 2×2). If $B = \begin{bmatrix} b & a \\ c & d \end{bmatrix}$ where $\det(A)$ represent determinant value of matrix A , then $\det B$ equals

Space for Rough Work

NAME OF THE CANDIDATE

FORM NO.

I have read all the instructions
and shall abide by them.

Signature of the Candidate

I have verified the identity, name and Form
number of the candidate, and that question
paper and ORS codes are the same.

Signature of the Invigilator

Space for Rough Work

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